

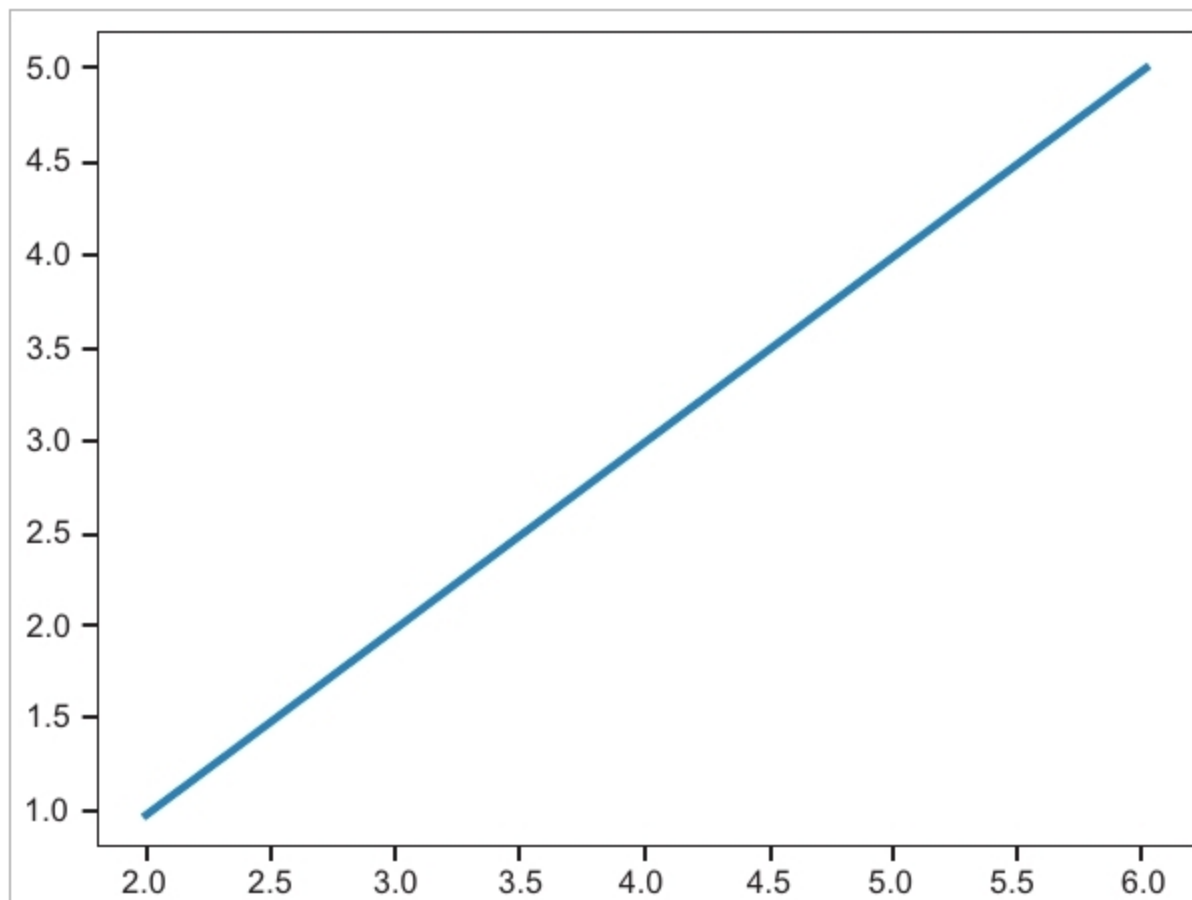


Figure 1: Plotting a line chart using Matplotlib

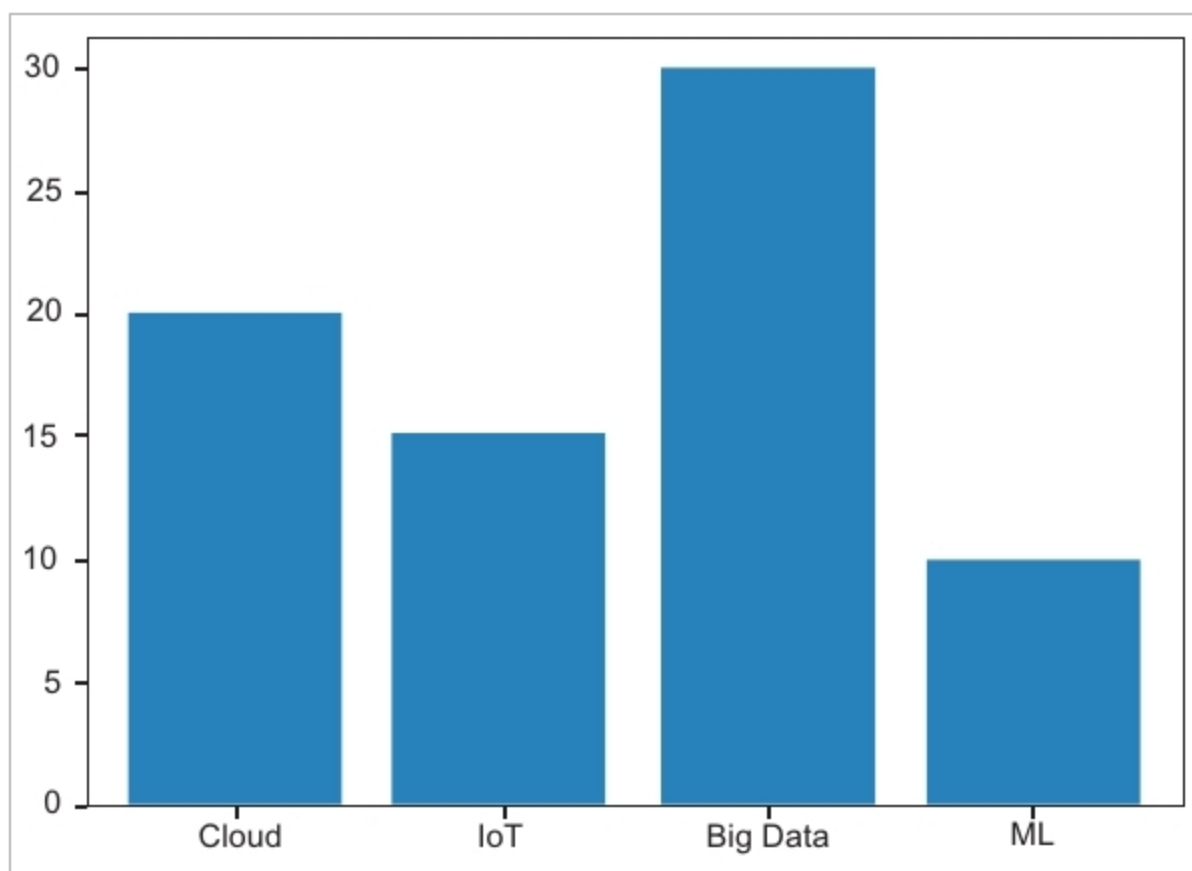


Figure 2: Plotting a bar chart using Matplotlib

capability of 3D and contour plots.

- *Geoplotlib* is an open source Python toolbox that serves to visualise geographical data. Its library supports the development of hardware-accelerated interactive visualisation and provides implementations of dot maps, kernel density estimation, spatial graphs and many other spatial visualisations.

Matplotlib is the mother of all Python libraries. It serves as an excellent base, enabling coders to wrap other tools over it. Now let's go into a detailed exploration of the Matplotlib data visualisation library and how it can be used to visualise various types of graphs.

## Different types of data visualisation plots

There are six key plots that you can use to understand data better. Each plot type is described below.

### 1) Visualisation using a line plot:

This is generally used to present observations collected at regular intervals. In Matplotlib, a line chart can be created by calling the plot method. The example below creates a sequence of even numbers on the X-axis and odd numbers on the Y-axis. The results are plotted as a line plot using the following Python code:

```
# importing the matplotlib library
import matplotlib.pyplot as plt
# the alias 'plt' has been set for
simplification purpose
# defining the dataset
x = [2, 4, 6]
y = [1, 3, 5]
#plotting the datapoints
plt.plot(x, y)
# display the plot
plt.show()
```

### 2) Visualising data with bar plots:

Bar plots show the distribution of data over several groups. A bar chart can be created using the bar

- capabilities of multiple Python APIs. *Seaborn* is built on top of Matplotlib and provides a simple interface for creating visualisations with a few lines of plotting codes. One of the benefits of using a Seaborn library is the ample choice of default styles and colour palettes, enabling coders to create

aesthetically pleasing and, most importantly, comprehensive chart designs.

- *Plotly* is a Web based toolkit for data visualisation. It is used to create visualisations that are highly interactive for the user. Plotly supports a variety of plots, right from a basic chart to the unique